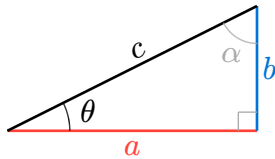


Trigonometric Formula Sheet

*Please check with your instructor before using this on an exam.



$$\theta + \alpha = 90^\circ$$

$$\text{area} = \frac{ab}{2}$$

$$a^2 + b^2 = c^2$$

$$a = \text{adjacent} = c \times \cos \theta$$

$$b = \text{opposite} = c \times \sin \theta$$

$$c = \text{hypotenuse} = \sqrt{a^2 + b^2}$$

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin \theta}{\cos \theta}$$

Pythagorean Identities

$$a^2 + b^2 = c^2 \Rightarrow \cos^2(\theta) + \sin^2(\theta) = 1$$

$$1 + \tan^2(\theta) = \sec^2(\theta) \quad \cot^2(\theta) + 1 = \csc^2(\theta)$$

Reciprocal Functions

$$\csc \theta = \frac{\text{hypotenuse}}{\text{opposite}} = \frac{1}{\sin \theta} \quad \sec \theta = \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{1}{\cos \theta}$$

$$\cot \theta = \frac{\text{adjacent}}{\text{opposite}} = \frac{1}{\tan \theta} \quad \sin^{-1}(\theta) \neq \frac{1}{\sin \theta}$$

Inverse Functions

$$\theta = \sin^{-1}\left(\frac{\text{opposite}}{\text{hypotenuse}}\right) = \arcsin\left(\frac{\text{opposite}}{\text{hypotenuse}}\right)$$

$$\theta = \cos^{-1}\left(\frac{\text{adjacent}}{\text{hypotenuse}}\right) = \arccos\left(\frac{\text{adjacent}}{\text{hypotenuse}}\right)$$

$$\theta = \tan^{-1}\left(\frac{\text{opposite}}{\text{adjacent}}\right) = \arctan\left(\frac{\text{opposite}}{\text{adjacent}}\right)$$

Function Domain Range

$\sin(\theta) = y$	θ in $(-\infty, \infty)$	y in $[-1, 1]$
$\cos(\theta) = x$	θ in $(-\infty, \infty)$	x in $[-1, 1]$
$\tan(\theta) = m$	$\cos \theta \neq 0$	m in $(-\infty, \infty)$
$\arcsin(y) = \theta$	y in $[-1, 1]$	θ in $[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\arccos(x) = \theta$	x in $[-1, 1]$	θ in $[0, \pi]$
$\arctan(m) = \theta$	m in $(-\infty, \infty)$	θ in $(-\frac{\pi}{2}, \frac{\pi}{2})$

Double Angle Identities

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$$

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2(\theta)}$$

Half Angle Identities

$$\sin\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

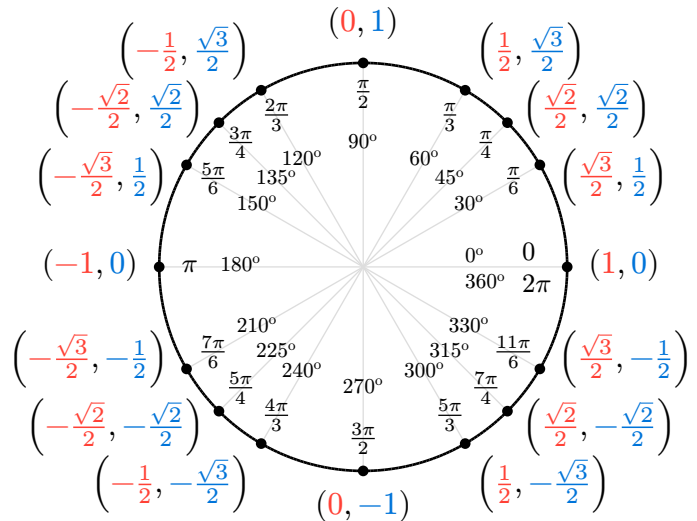
$$\tan\left(\frac{\theta}{2}\right) = \frac{\sin \theta}{1 + \cos \theta} = \frac{1 - \cos \theta}{\sin \theta}$$

Power Reduction Formulas

$$\sin^2(\theta) = \frac{1}{2}(1 - \cos(2\theta)) \quad \cos^2(\theta) = \frac{1}{2}(1 + \cos(2\theta))$$

$$\tan^2(\theta) = \frac{1 - \cos(2\theta)}{1 + \cos(2\theta)} \quad \sin \theta \cos \theta = \frac{1}{2} \sin(2\theta)$$

The Unit Circle ($\cos \theta, \sin \theta$)



$$\text{pattern: } \frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$$

$$\text{rad} = \text{deg} \times \frac{\pi}{180^\circ}$$

$$\theta \pm 2\pi = \theta$$

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = -\tan \theta$$

$$\sin(\theta \pm \pi) = -\sin \theta$$

$$\cos(\theta \pm \pi) = -\cos \theta$$

$$\tan(\theta \pm \pi) = \tan \theta$$

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta$$

Sum Identities

$$\sin(\theta + \varphi) = \sin(\theta) \cos(\varphi) + \cos(\theta) \sin(\varphi)$$

$$\cos(\theta + \varphi) = \cos(\theta) \cos(\varphi) - \sin(\theta) \sin(\varphi)$$

$$\tan(\theta + \varphi) = \frac{\tan \theta + \tan \varphi}{1 - \tan(\theta) \tan(\varphi)}$$

Product-to-Sum Formulas

$$\sin(\theta) \sin(\varphi) = \frac{1}{2}(\cos(\theta - \varphi) - \cos(\theta + \varphi))$$

$$\cos(\theta) \cos(\varphi) = \frac{1}{2}(\cos(\theta - \varphi) + \cos(\theta + \varphi))$$

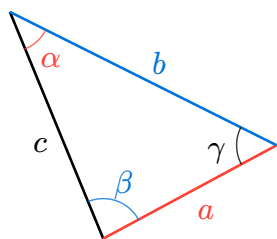
$$\sin(\theta) \cos(\varphi) = \frac{1}{2}(\sin(\theta - \varphi) + \sin(\theta + \varphi))$$

Sum-to-Product Formulas

$$\sin \theta + \sin \varphi = 2 \sin\left(\frac{\theta + \varphi}{2}\right) \cos\left(\frac{\theta - \varphi}{2}\right)$$

$$\cos \theta + \cos \varphi = 2 \cos\left(\frac{\theta + \varphi}{2}\right) \cos\left(\frac{\theta - \varphi}{2}\right)$$

$$\cos \theta - \cos \varphi = -2 \sin\left(\frac{\theta + \varphi}{2}\right) \sin\left(\frac{\theta - \varphi}{2}\right)$$



The Law of Sines

$$\frac{a}{\sin(\alpha)} = \frac{b}{\sin(\beta)} = \frac{c}{\sin(\gamma)} = \frac{abc}{2 \times \text{area}}$$

The Law of Cosines

$$a^2 = b^2 + c^2 - 2bc \cos(\alpha)$$

The Law of Tangents

$$\frac{a-b}{a+b} = \frac{\tan\left(\frac{\alpha-\beta}{2}\right)}{\tan\left(\frac{\alpha+\beta}{2}\right)}$$

$$\alpha + \beta + \gamma = 180^\circ$$

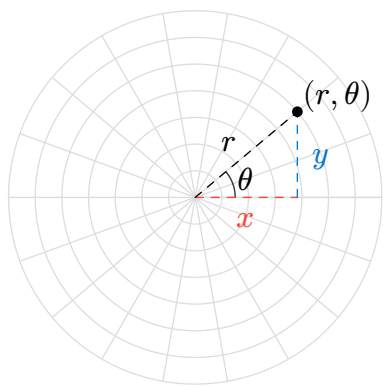
Polar Coordinates

$$x = r \cos \theta$$

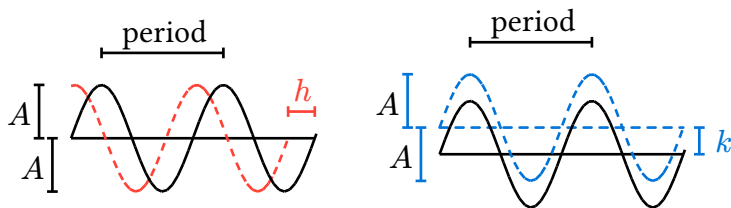
$$y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \arctan\left(\frac{y}{x}\right)$$



Waves



$$f(x) = A \sin(\omega x - \varphi) + k = A \sin(\omega(x - h)) + k$$

A = amplitude

$\varphi = \omega h$ = phase shift

ω = angular frequency

$h = \frac{\varphi}{\omega}$ = horizontal shift

$$\text{period} = \frac{2\pi}{\omega} = \frac{1}{\text{frequency}}$$

k = vertical shift

$$\text{frequency} = \frac{\omega}{2\pi} = \frac{1}{\text{period}}$$

Polar Coordinates and the Complex Plane

$$x + iy = r(\cos \theta + i \sin \theta)$$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

for any integer n :

$$(r(\cos \theta + i \sin \theta))^n = r^n(\cos(n\theta) + i \sin(n\theta))$$

Trigonometry in Calculus

Derivatives

$$\sin'(x) = \cos x \quad \cos'(x) = -\sin x$$

$$\tan'(x) = \sec^2(x) \quad \cot'(x) = -\csc^2(x)$$

$$\sec'(x) = \sec(x) \tan(x)$$

$$\csc'(x) = -\csc(x) \cos(x)$$

$$\arcsin'(x) = \frac{1}{\sqrt{1-x^2}} = -\arccos'(x)$$

$$\arctan'(x) = \frac{1}{1+x^2} = -\text{arccot}'(x)$$

$$\text{arcsec}'(x) = \frac{1}{|x| \sqrt{x^2-1}} = -\text{arccsc}'(x)$$

Antiderivatives

$$\int \sin(x) dx = -\cos x + C$$

$$\int \cos(x) dx = \sin x + C$$

$$\int \tan(x) dx = \ln|\sec x| + C$$

$$\int \sec(x) dx = \ln|\sec x + \tan x| + C$$

$$\int \cot(x) dx = -\ln|\csc x| + C$$

$$\int \csc(x) dx = -\ln|\csc x + \cot x| + C$$

Taylor Series

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

$$\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$$

Trig Substitutions

$$\sqrt{a+x^2} \rightarrow \text{try } x = \sqrt{a} \tan \theta$$

$$\sqrt{a-x^2} \rightarrow \text{try } x = \sqrt{a} \sin \theta$$

$$\sqrt{x^2-a} \rightarrow \text{try } x = \sqrt{a} \sec \theta$$

Hyperbolic Trig

$$\sinh(x) = \frac{1}{2}(e^x - e^{-x})$$

$$\cosh(x) = \frac{1}{2}(e^x + e^{-x})$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)}$$

$$\sinh^2(x) + \cosh^2(x) = 1$$

$$\sinh'(x) = \cosh(x)$$

$$\cosh'(x) = \sinh(x)$$

Cylindrical Coordinates

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$dxdydz = (r)drd\theta dz$$

Spherical Coordinates

$$x = \rho \cos(\theta) \sin(\varphi)$$

$$y = \rho \sin(\theta) \sin(\varphi)$$

$$z = \rho \cos \varphi$$

$$dxdydz = (\rho^2 \sin \varphi)drd\theta d\varphi$$

Specific Trig Integrals

$$\int \sin^m(\theta) \cos^n(\theta) d\theta :$$

$$m \text{ odd} \rightarrow -\int f(\cos \theta)(-\sin \theta) d\theta$$

$$n \text{ odd} \rightarrow \int f(\sin \theta) \cos(\theta) d\theta$$

otherwise use power reductions

or integrate by parts

$$\int \tan^m(\theta) \sec^n(\theta) d\theta :$$

$$m \text{ odd} \rightarrow \int f(\sec \theta) \tan(\theta) \sec(\theta) d\theta$$

$$n \text{ even} \rightarrow \int f(\tan \theta) \sec^2(\theta) d\theta$$

(applies with cot & csc too)

